

Generative AI for Social Welfare Access: A Case Study of a Multilingual *Bürgergeld* Application Chatbot in Germany

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1. Executive Summary

Germany's complex asylum and social welfare system can create significant challenges for asylum seekers and refugees in accessing essential benefits like *Bürgergeld* (unemployment benefits for those holding residence permits, refugee protection status or Ukrainians arriving in Germany before April 1st 2025). Since 2015 this has further been elevated by severe administrative backlogs that slow down case workers, and language barriers that prevent the seekers of such benefits from easily completing required paperwork.

The following report examines the case of a GenAI chatbot deployed in Germany, and developed for a social welfare NGO, to guide non-German speaking applicants in completing their application forms. The chatbot essentially acts as an intelligent data collection interface, and helps caseworkers with the generation of an application that they can then review. The system has received positive feedback from caseworkers in the NGO for the alleviation of workload and assistance with difficult tasks, as GenAI tools often do when deployed for particular use cases across the public sector. However, it has also posed challenges for data protection and digital sovereignty, and has raised concerns of potential biases, such as linguistic and automation bias. Although the system did not eventually scale, it serves as an important use case upon which this report sheds a critical lens and gives recommendations to improve transparency and accountability that future GenAI deployments should keep note of.

2. Generative AI in the Public Sector

The use of generative AI (GenAI) and Large Language Models (LLMs) represents a critical shift in public administration, which redefines the relationship between government services and citizens that use them. Public sector entities globally are shifting towards increased adoption, with the OECD highlighting that 67% of countries are utilising AI to enhance public service design and delivery, aiming to improve efficiency and address growing workloads (OECD, 2025). This surge is partly due to GenAI's ability to automate high-volume administrative tasks, which allows civil servants to focus on the casework itself. BCG has estimated that productivity gains for the public sector from the use of GenAI can be valued at 1.75 trillion USD by 2033 (Carrasco et al., 2023). This use, however, brings its own set of challenges.

Over the past semester, throughout this course, we have explored frameworks and case studies to gain a deeper understanding of what the deployment of AI in the public sector entails and how to effectively assess its implementation and integration through various critical lenses. We analysed the different narratives surrounding the use of AI in the public sector, both positive, transformative ones, and negative, existential risk views, and examined core ethical and governance principles, such as transparency, explainability, fairness, and the resulting equality of treatment when using AI. Through reflections on various systems in place and the changes some of these systems underwent, the importance of carefully integrating such systems became clear, despite their helpfulness, across public sector processes, especially when affecting critical services that touch on fundamental individual rights and public trust.

Accordingly, in the following report, we will explore a specific German case in which a GenAI chatbot was used to assist asylum seekers and recognised refugees with navigating complex information on *Bürgergeld* benefits and applications. This initiative offers a unique perspective on the intersection of German and EU law (e.g., GDPR), humanitarian needs, and recent technological developments. By examining its development, technical architecture, and implementation challenges, we can better understand how its design relates to the principles of transparency, fairness, and equality of treatment within a vulnerable population, and what recommendations we can keep in mind moving forwards

3. Use of generative AI in Asylum & Social Security in Germany

Germany has been a major destination for asylum seekers and refugees ever since the 2015 refugee crisis. Historically, its approach to refugee laws was primarily focused on assisting displaced persons and war survivors of German origin (i.e. *Kriegsvertriebene*, [Bundesvertriebenengesetz](#) (BVFG)), favouring a clear administrative and bureaucratic

approach, often easily accessible for German speakers. Over the following decades, Germany's asylum framework became increasingly shaped by European Union regulations, particularly the Dublin Regulation in the 1990s and 2000s, which aimed to harmonise asylum procedures across EU member states ([BPB](#), 2018). The country provides social welfare support, such as asylum seeker benefits (defined in the [Asylbewerberleistungsgesetz](#), providing up to 441 Euro monthly), or in specific cases *Bürgergeld* (unemployment benefits for those holding residence permits, protection status or Ukrainians arriving in Germany before April 1st 2025).

In the aftermath of the sudden influx of asylum seekers from Syria, Iraq, Afghanistan, and other countries starting in 2015, the enormous strain on Germany's administrative capacities became apparent. Processing times lengthened, and the complexity of forms, interviews, and documentation requirements highlighted persistent bureaucratic bottlenecks. Today, the *Bundesamt für Migration und Flüchtlinge* ([BAMF](#), Federal Office for Migration and Refugees) is responsible for handling asylum claims, but despite modernisation efforts, many applicants still experience communication challenges. This is particularly due to language barriers and the intricacies of German administrative procedures. To this end, the following section will assess Germany's unique challenge of digitising its humanitarian aid, while also examining how algorithmic systems could facilitate processing and/or infringe upon the German Basic Law (*Grundgesetz*). Due to Germany's federal structure, digital solutions have been slow to be implemented; overall, worsened by fragmented state-level processes, stringent data protection rules, and the need for human oversight in sensitive decisions.

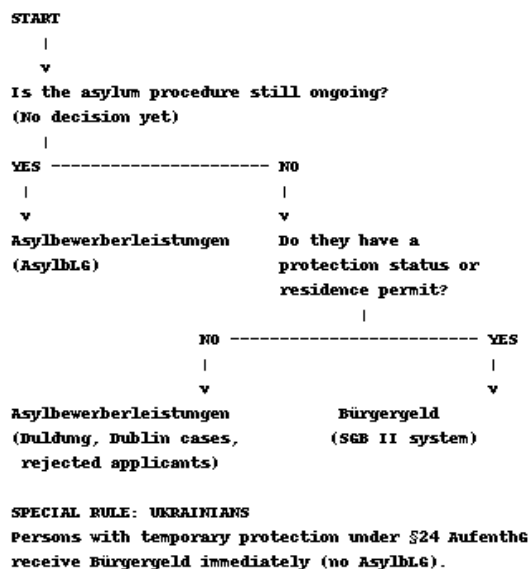
3.2 Legal Basis of Algorithm Use in Humanitarian and Social Aid

The allocation of social welfare in Germany is based on a multi-level legal framework, which also applies to the use of algorithms and AI in humanitarian aid. Created in response to the horrors of the Second World War, it aims to protect individuals' rights and ensure fair allocation. At the European Union level, the General Data Protection Regulation ([GDPR](#)) governs the protection of personal data, including that of refugees and asylum seekers. The GDPR, along with the AI Act, places particular restrictions on automated decision-making, requiring transparency, accountability, and the possibility for human intervention in particular when individual rights could be affected.

The German *Grundgesetz* also guarantees fundamental rights that intersect with algorithmic use, including equality before the law (Art. 3 [GG](#)), the right to human dignity "*Menschenwürde*" (Art. 1 [GG](#)), and protection of personal data under the general right of personality (Art. 2 [GG](#) in conjunction with Art. 1 [GG](#)). Any use of AI in the public sector must respect these constitutional protections, particularly for vulnerable populations such as

refugees and asylum seekers.

At the national level, the German Social Code ([SGB II](#)) regulates social welfare benefits, including social benefits for asylum seekers and certain refugees during the early stages of their application or temporary residence ([Asylbewerberleistungsgesetz. AsylbLG](#)), as well as *Bürgergeld*. It outlines clear documentation and eligibility requirements (§ 1 AsylbLG Leistungsberechtigte (entitled persons) & § 2 ff. Types of benefits and conditions): ¹



According to **German Social Law** ([SGB I](#) & [SGB II](#)) (§31 [SGB X](#)), decisions must be transparent, reasoned, and subject to challenge. Therefore, they require human oversight. While algorithms and AI systems can assist in administrative tasks (verifying documents, detecting inconsistencies, or facilitating communication), they cannot make legally binding decisions regarding eligibility or benefit allocation without human oversight (§35 [VwVfG](#), §24 [VwVfG](#)). These rules precede the GDPR Art. 22, but are already mirrored in the [Grundgesetz \(Art. 1 & Art. 19\(4\)\)](#).

3.3 Germany's Struggle with Digitisation in Humanitarian Aid

Keeping these facts in mind, the potential of digital tools to streamline asylum procedures and humanitarian aid has faced persistent challenges in digitising Germany's refugee support system. Germany requires a close documentation system within the local municipalities (*Gemeinde*) and the State (*Bundesland*); asylum and social welfare administration are split between federal, state, and municipal authorities, each with its own processes and IT systems. The sensitivity of personal refugee data further complicates digitisation. Refugees' information includes not only identification and residency details but

¹ Generated with the help of <https://codetoflow.com>, using a text code generated on Google Collab.

also health records, family composition, and previous migration history. Applicants will have to interact with multiple actors such as the *BAMF*, *Ausländerbehörde* (registration office for foreign nationals (outside of EU)), *Jobcenter*, *Gesundheitsamt* (health office) and the *Sozialamt* (welfare office) ([BMI](#)). Within these institutions, paper-based workflows remain common, especially in smaller municipalities or local offices, where forms are often manually processed. This is also due to the high cost of digital solutions, which must be audited and comply with legal requirements (Cf. Legal background).

The BAMF has implemented some tools to accelerate federal processes, which favour public state actors. They are not specifically designed to help refugees navigate this system. It includes [FLORA](#) (“*Föderale Blockchain Infrastruktur Asyl*”), and the [digitisation of the application process, eg. DIAS](#) (“Language Biometrics Assistance System”, to investigate an undocumented refugee’s region of origin through language patterns). On the applicants’ side, there are only a few initiatives. Examples include the [Integreat](#) App, containing local information for refugees (registering, health, employment, education) in various languages, as well as an [AI chatbot](#) for the general population by a welfare NGO that informs individuals about eligibility for *Bürgergeld* (Neue Caritas, 2024). The case study Chatbot in this report that fills out *Bürgergeld* forms in the applicant’s native language thus fills an important void on the applicants’ side, as they have few tools to navigate this complex legal and bureaucratic system. While the laws in place aim to protect individual rights and hold institutions accountable, they are not designed to process large numbers of applications. The system must either reform (which is unlikely due to the supremacy of EU law and Basic Law) or build transparent and lawful technological solutions.

3.4 Implications for algorithmic systems

Out of Germany’s legal tradition following principles emerge as a paramount guideline. Since refugees and asylum seekers are a vulnerable group, even after their decision of asylum has been positive, we may derive said doctrines for the technology’s design:

1. Fairness and non-discrimination: no individual is disadvantaged based on gender, ethnicity, nationality, or other sensitive attributes (Art. 3 [GG](#)).
2. Transparency and accountability ([SGB](#) X §§33–35): “no black boxes”.
3. Auditing for biases.
4. Human oversight ([GDPR](#) Art. 22 *VwVfG*).

These lead to guiding principles of AI design in public administrations, aligning with EU law. Opposition to the use of algorithms in refugee contexts often arises from civil society and activist concerns, particularly the risks of profiling, discrimination, and increased surveillance. There is significant scepticism regarding the use of AI in sensitive decision-making scenarios, as many question whether automated systems can adequately account for

complex humanitarian needs. It also echoes points raised by the [Deutscher Ethikrat](#), an oversight committee for moral guidance of public debates and policies, which recommends certain principles based on Germany's legal framework, especially when dealing with vulnerable populations.

- Human-in-the-loop principle mandatory. [GDPR Art. 22](#); [Deutsche Ethikrat](#); [EU AI Act](#).
- Consent and data minimisation are essential under [GDPR Art. 5–7](#); EU Charter Art. 8.
- Documentation and auditability: authorities must justify decisions made with algorithmic support. [EU AI Act](#); [GDPR accountability principle \(Art. 5\(2\)\)](#).

AI in Germany's humanitarian aid, therefore, echoes most of its European neighbours. German state actors will have to assess such tools in the future to fulfil their legal humanitarian obligations, which, due to the slow processing time, might infringe on individual rights and basic law. This report, therefore, also raises questions about when technological solutions and/or administrative innovations become a moral and legal obligation.

4. Case Study and Analysis²

The reported system is a generative AI-based chatbot interface that assists asylum seekers and refugees in filling out forms for Bürgergeld (unemployment benefits) in their native language, provided by a social welfare organisation (NGO).

4.1 System Background and Stakeout Process

Background & Context

The chatbot emerged from a practical, ground-level discussion. The developer, when discussing the organisational processes of a German welfare association with an associate, wanted to address the pain points impacting case workers' jobs. Frontline caseworkers were overwhelmed by the effort required to assist asylum-seeking clients in completing the foundational benefits application in German. Although few translated versions of the form existed, they were mainly convoluted and inconsistently updated, making it challenging for clients with limited German proficiency to complete. Caseworkers were repeatedly guiding clients through the same question forms and manually translating the same bureaucratic language. The pain point was not a lack of legal expertise, but rather administrative overload.

Casual conversation, rather than a formal policy initiative, was the starting point for

² The information presented in this section is based on an interview with one of the developers involved in the chatbot's creation. For reasons of confidentiality and anonymity, no identifying details about the developer or the NGO that requested the tool can be disclosed. More information regarding this can be found in the annexe.

this AI tool. The developer, with their interdisciplinary background, posed the question to his associate: could a chatbot handle the form-filling dialogue in a clearer, multilingual way? This engendered the prototype's origin as the two associates experimented with a standard chatbot interface, prompting it to fill out the *Bürgergeld* form with statements such as 'I don't speak German. Please fill out the form in language [X].' The promising results led the developer to build a dedicated prototype, which was then presented to the association's caseworkers. This had a positive response, with much enthusiasm, which encouraged the continuation of the development.

Large language models, particularly ChatGPT and Claude, played a notable role in shaping the system. The developer referred to them for conceptual troubleshooting, design advice and brainstorming technical approaches. In Winter 2024, the AI system was rolled out. The three-month concept rollout timeline followed an iterative process involving recurring testing and refinement with two test users, including an Afghan and a Ukrainian applicant. Later, a broader group of clients tested the product, representing linguistic backgrounds including Russian, one Arabic dialect, Farsi, Pashto, Urdu, French, and English.

Stakeout Process

The stakeout phase, which involved identifying the problem, validating user needs, and iterating solutions, was deeply user-driven. Input came from caseworkers and, on a secondary level, from clients who provided feedback on clarity and ease of use. No complaints were given about language accuracy. Caseworkers were especially supportive, as unlike top-down digitalisation strategies, this small-scale system addressed a concrete, daily pain point in simplifying multilingual communication and capturing information in a structured format for later review.

The developer prioritised capability and multilingual reliability when choosing models. Although they explored open-source models extensively, their paper performance did not translate well to real-world stability. Many models were too large to run locally and required third-party US hosting, which raised concerns for compliance. However, the system itself was hosted on the German provider 1&1 Cloud. Proprietary models, such as Gemini and OpenAI's GPT 4.1, offered the best combination of accuracy, following instructions, multilingual performance, and permissible usage through paid API access. The system stored no conversational data locally beyond structured field values and minimal metadata, while API providers retained logs for up to 30 days in accordance with their standard privacy policies.

The conversational flow was controlled by a combination of LLM guidance and hard-coded logic. The chatbot asked questions, recorded responses in a structured

database, and triggered a second LLM process to check for inconsistencies. Once the system deemed the form complete, the conversation ended, and a final PDF-style report was generated for caseworkers. Importantly, the system never made legal or eligibility decisions. It acted as an intelligent data collection interface. Human oversight occurred entirely at the end, when a caseworker reviewed the generated form and assisted the applicant with the official submission.

Why Did the System Not Scale Up? Challenges of Development

Despite early functional success, the chatbot did not scale within the organisation. This was partly due to legal uncertainty, as Germany's legal system restricted who could provide legal advice. Even though the chatbot was framed as an administrative tool, expanding it would require formal organisational endorsement and more scrutiny around privacy, liability, and legal compliance. Questions created uncertainties, such as how to ensure GDPR compliance beyond API-level guarantees or how to verify that outputs could never be construed as legal advice. These concerns were manageable at a small scale, but proved to be barriers to institutionalising the tool.

There were uncertainties around data privacy as the organisation was uncertain about sending sensitive personal data to US-based proprietary models, even via secure APIs. While they ensured the system encrypted data in transit and stored minimal information, the reliance on external providers remained a sticking point. While the tool cost only around 10–50 cents per conversation, scaling required formal procurement channels, stable hosting arrangements, and organisational IT approvals. The organisation lacked rapid-deployment structures comparable to those found in private-sector innovation units (e.g., DARPA environments). Without such structures, the path to institutionalisation was cumbersome. Anecdotally, the tool seemed helpful, but the NGO eventually discontinued the specific *Bürgergeld* service for which the chatbot had been developed. Without that service, the tool's core use case disappeared. It remained unclear whether the chatbot reduced staff workload enough to justify a long-term operational mandate.

The developer pointed to a broader pattern that explains this, where public sector environments often lack interdisciplinary teams capable of rapid experimentation. While individual innovators can create prototypes, sustaining and scaling them requires organisational alignment, legal guidance, IT infrastructure, and governance structures that many institutions simply lack.

4.2 Description of the system

How the Chatbot Functions

The chatbot's interface understood input from clients in their native languages, generated outputs to collect key information, and finally produced a report for the case workers. To

achieve this, the system integrated natural language understanding, conversational flow management, and dynamic response generation in a chatbot built on top of Gemini and ChatGPT. The system's operational flow was as follows:

1. The client approaches the organisation to request support in applying for social benefits, specifically *Bürgergeld*.
2. Later, the case is assigned to a social worker, who generates a new procedure in the system and sends the client the link to the chatbot.
3. The client receives the link and replies to it in their own time, using their smartphone or computer, as internet access is required.
4. The client interacts with the chatbot using an interface for answering questions. They first select the language of the conversation with the chatbot. Then, the chatbot asks questions related to the fields it must fill. At any time, the client can update the information provided to the chatbot or request that it delete the conversation; in this case, the conversation will be removed from the organisation's database.
5. In parallel, the chatbot has different options in the conversation:
 - a. Provide a text that fosters the conversation (for this Gemini is used).
 - b. Record the information provided by the client in their database, as it is a required field in the application form (using Gemini).
 - c. A second LLM processes this conversation to check for inconsistencies (still using Gemini).
 - d. Conduct in-depth research on a specific issue using ChatGPT. For example, if the chatbot in the conversation asked, "What is your marital status?" and the client then follows up with "I am not sure, I got married in Kabul at 16, but I am not sure that it is recognised." The chatbot will review preloaded documents relevant to the case or search the internet for the response.
6. When the chatbot has gathered all the necessary information, it thanks the client for their interaction and forwards the information to the case worker. The output generated by the application is a final report, which is an online version of the dataset generated while the chatbot was gathering information. This report allows the case worker to view the final report in German, as well as the input that generates the report (title, record answer, original answer), and the conversation flows between the client and the chatbot in the original language.
7. The case worker receives the information and schedules a meeting with the client. In that meeting, both submitted the application for *Bürgergeld*, along with the provided information.

Organisational Impact & Oversight

The initial reaction from the caseworkers was enthusiastic; they drew on their own experience and found it incredible how much potential the system had for their day-to-day work. Despite never meeting the caseworkers directly, the developer was told that the reaction was highly positive. Such a conclusion is reasonable, as prior to the chatbot's deployment, as outlined above, the process had been highly bureaucratic and linguistically demanding.

Consistent with its design, the chatbot did not make any legal or eligibility decisions; caseworkers remained fully responsible for reviewing and submitting applications. It only gathers clients' personal information. This allows the case worker to revert to the “manual” method of filling out the *Bürgergeld* application when the information gathered by the chatbot appears incorrect. They can then seek assistance from a coworker to translate the client's response or use an electronic system to directly engage with them.

4.3 Analysis of the system***Data protection & Legal concerns***

The main risks of the system are the data protection risks associated with using the “pro” versions of proprietary US-American LLMs. As stated above, the system stores no conversational data locally beyond structured field values and minimal metadata, and is hosted in Germany, but the LLM API provider companies retain logs for up to 30 days for compliance reasons. These data transfer risks are significant, and for digital sovereignty reasons, exploring the Mistral API might have been an opportunity for improvement, considering Mistral is building up their own computing infrastructure, Mistral Compute (The Outpost, 2025). In reality, however, value for money in an already underfunded sector was the overarching concern and primary driver for choosing Google's Gemini API (and OpenAI's GPT-4.1) in a pilot project based on this type of volunteer cooperation.

Other legal uncertainties arose mostly in the context of compliance with specific sectoral legislation beyond the GDPR, and part of the reason why scaling was considered challenging (see *Stakeout process and Privacy & Security Risks*).

Policy Impact & Access

Subsequently, we asked the developer about the policy impact of the system, including improvements in access to social security benefits, as well as revelations about the policies' complexity.

Concerning the question if the system changed who applies for *Bürgergeld* (such as demographic shifts or reduced barriers for specific communities (language groups, elderly, etc.)), the developers said that he did not receive any direct feedback from his organisation on this, but believe there no be no direct impact on expanding the types of people that apply

since the system is not publically accessible. The system is only accessible to clients who are already registered, with the link to the interface generated by the caseworker via phone or computer. However, speeding up the process and supporting the most challenging step likely freed up capacities among the caseworkers, potentially allowing them to serve more clients. This is also supported by the overall organisation feedback that the system helped to manage the workload. The system thus contributes to improved access to social security benefits for refugees, albeit through the mediation of caseworker capacities.

Governance, Transparency & Accountability

Questions surrounding governance, transparency & accountability of systems are at the heart of the organisational, political and technical interactions between the system and the public. This is why we explored the questions of public awareness and controversy, oversight structures, contestability and responsibility in depth. The overall public is not aware of the tool, as it was only used internally in one of the organisation's many chapters in Germany, and thus there has been no public backlash. There was also no promotion within the organisation or external guidance from other organisations. However, plans were made to expand the system to more chapters after one year in deployment, which were abandoned for the reasons in the *stakeout process* section.

Regarding the governance structure for oversight and contestability of decisions influenced by the chatbot, the system assigns clear responsibilities to the caseworker. The developer describes the system as a “fancy Google form” that the caseworker receives; however, it takes all decisions on how to move forward, e.g., with flags together with their client, before submitting the formal request. Documentation and transparency mechanisms, such as impact assessments and audit reports, were not done, but a general privacy policy and terms of service drafted by the developer and their contact at the organisation, also with the help of AI, exist. Users can contest the Chatbot's outputs and inputs in various ways: they are informed which information is input into the form fields, and can demand rectification in the conversation, as well as complete deletion of the conversation. The developer states that there were no instances where the form was not submitted to the caseworker. If the outputs nonetheless do not align with the desired inputs, they can be discussed with the caseworker during a conversation, where either a translator is present or other AI-based translation tools are utilised. The primary limitation of contestability is thus the language capabilities of the AI models used. The official application for social benefits is only filed when any inconsistencies and disagreements between the client and caseworker have been resolved

Similarly, if the chatbot provides incorrect information that harms an applicant, e.g. on the validity of an Afghan marital status acquired at 16, the caseworker, as an intermediary

between the client and the state, is always responsible for making sure no (false) information that could harm the client is retained or influences the final request for social benefits. Through the overview, they are sent from the conversation (including the entire conversation in the client's language, the inputs in the client's language, and the German outputs), they can verify and contest, together with the client, the origin of a potential flag or false information. The system thus only prepares the information for the caseworker and flags potential inconsistencies, missing inputs, etc. While the system does not recommend singular decisions, it does, however, make decisions in many small steps on what to input into the fields and what to leave out. This might nonetheless contribute to some automation bias, especially considering that it is not framed as an automated decision-making system. This is why we recommend displaying the entire conversation in German, as well as in the native language, so caseworkers can transparently verify which parts of the conversation were input into the form and discuss why.

The developer and the organisation itself (besides the caseworker team leader) are not directly involved in oversight and accountability, as, for example, the organisation's chapters act predominantly independently of each other. The developer nevertheless acts upon issues raised by the caseworkers to change certain features, which happened iteratively during the three-month pilot. The developer was also not remunerated for any of their work, but did this as a volunteer side project.

The system was in use from November 2024 to summer 2025, after which the developer was informed that the organisation had stopped using the system. They did not receive any feedback about complaints, etc., but were informed that the organisation overall scaled back their help for refugees, including access to social benefits. This is likely connected to the overall changing discourse surrounding refugees in Germany as well as the Federal Government's decision in summer 2025 to end the privileged position for new Ukrainian refugees, who, ever since the start of the war have all received *Bürgergeld* under the temporary protection status. The system itself thus does not seem to have caused its own discontinuation but rather the overall political situation.

To conclude, the system overall strikes a good balance between automatising tasks that the organisation does not have enough resources and skills to do, yet implementing meaningful human oversight by leaving contestation, discretion and responsibility with the subsequent caseworker-client personal interaction.

5. System Risks & Challenges

While the system generally strikes a good balance between improving efficiency and workload and human oversight, some risks inherent in the use of generative AI in the public sector and the field of asylum and social security applications specifically persist.

5.1 Privacy & Security Risks

Firstly, implementing generative AI chatbots into the administrative process for applying for Bürgergeld benefits requires examination and information of security and privacy risks. This is essential, as these values help alleviate users' concerns about potential abuse or mishandling, thereby facilitating the implementation process (Abdenebaoui et al., 2025). Naturally, in order to determine eligibility, the Bürgergeld application process requires the input of sensitive personal information. Traditionally, these applications are filled out manually in person or online, with the assistance of an interpreter for individuals who do not speak German sufficiently (Arbeitsagentur, 2023). Here, the introduction of a generative AI system exposes an additional vulnerability. In terms of language accessibility, the main objective of this chatbot is to facilitate the application process for a diverse range of multilingual users. Thus, it functions best if built on top of frontier GPAI models such as Gemini or ChatGPT. When users interact with the chatbot, their sensitive information is routed to the company's servers, where it is processed and stored. Yet, the details about the practices that developers use to safeguard users' privacy remain obscure (Itoi, 2025). OpenAI (2025) officially states that, for business-paid customers, the data is not used for training by default. Nonetheless, the company has a colourful history of security breaches and privacy scandals. These include hackers extracting personal information from the training dataset (White, 2023), being fined 15 million euros for training OpenAI models using personal data without an appropriate legal basis (Zampano, 2024), and credential leaks for API accounts as recently as November 2025 (Wells, 2025). In this respect, data breaches are particularly concerning because the questionnaires contain sensitive information regarding a person's family, employment, finances, bank details, and migration and medical history. Thus, data breaches and unauthorised access could not only undermine the trust in the system (Abdenebaoui et al., 2025) but might also present an immediate threat to applicants. In extreme cases of political refugees, this could lead to re-identification, localisation, intimidation, or even potential violence by malicious actors. Thus, there appears to be significant tension between achieving efficient multilingual performance of the chatbot while safeguarding users' privacy and security.

5.2 Procedural Fairness

Naturally, any conversation about the potential risks associated with implementing generative AI in administrative procedures, however remote, would not be complete without discussing procedural fairness. Although the chatbot does not directly determine application outcomes, it is crucial to examine its role within the application process. The integration of the chatbot into the administrative process is accompanied by several safeguards designed to promote transparency, explainability, contestability, justifiability and accountability. While

these efforts reduce the main risks, they do not eliminate them entirely. Regarding contestability, users can freely ask the chatbot to update, change, or delete any information, granting them a high degree of autonomy over content. Regarding accountability, the full application is reviewed by the caseworker in collaboration with the refugee, ensuring human oversight and personal responsibility. However, transparency, justifiability and explainability remain only partially addressed. The organisation attempts to promote transparency and traceability by providing caseworkers with a complete transcript of user interactions with the chatbot. This approach, however, does not resolve language barriers. Caseworkers may receive transcripts in languages they do not understand, preventing them from verifying system outputs. Similarly, refugees may access the full application in German but may lack the language proficiency to understand and contest errors. Consequently, the process relies heavily on the accuracy of AI-assisted translation and interpretation. The outputs (generated based on the application of the hardcoded logic / symbolic AI that processes the conversational input) are also not explicitly justified to the user and caseworker. The caseworker can only infer from the conversation and inputs if they understand the clients' native language (and vice versa). Providing the caseworker with an additional translation of the conversation would help here, as the translation of the inputted and outputted form should already be "identical". Additionally, the system cross-references user inputs with relevant legislation to complete the questionnaire. For instance, it may search for laws from the refugee's home country to determine their marital status or financial situation. These determinations depend on precise terminology and cultural context, which general-purpose chatbots may not accurately interpret, increasing the risk of errors. In certain situations, the system makes decisions effectively by inputting information into the form. This dynamic is problematic because interaction with AI systems may create an automation bias where humans uncritically accept AI outputs rather than overseeing them. This is likely to be worsened by the overwhelming workload of caseworkers, which the system was designed to alleviate. Under time constraints, caseworkers might not have sufficient capacity to inspect the full transcript for each refugee and cross-reference system outputs, further diminishing explainability and transparency.

5.3 Language Bias

Lastly, the introduction of generative AI into the application process may introduce hidden linguistic biases. As users interact with the chatbot in their native language, the performance of the AI system relies on the natural language processing capabilities of the underlying AI model. In turn, the usefulness of these large language models depends on the quality of the underlying dataset on which they were trained. As most frontier LLM models are trained in English, they are not accustomed to the numerous languages from the Global South (Pava

et al., 2025). Due to a general scarcity of data, both in terms of quality and quantity, the most popular models tend to underperform when applied to non-English contexts (ibid.). In the context of refugees, this language bias poses a significant obstacle as applicants arrive from a myriad of countries, each with vastly different languages and dialects. In Germany, the overwhelming majority of refugees come from the Global South, with countries such as Syria, Afghanistan, Iraq, Somalia, and Iran being among the most represented (AIDA, 2025). Thus, as most of the users are likely to interact with the chatbot in non-English languages, they might be disproportionately disadvantaged by the adoption of the chatbot into the Bürgergeld application process.

Furthermore, the linguistic bias becomes amplified when designing LLM-based applications for low-resource languages in contexts that require a high degree of linguistic accuracy and subject-specific expertise (Tassios et al., 2025). Looking at the institutional context, German administrative procedures are characterised by their complexity, opacity, and frequent reliance on professional jargon (Aljuneidi et al., 2025). As mentioned earlier, frontier general-purpose models often lack the syntactical and cultural fine-tuning to provide reliable foreign language outputs in complex contexts (ibid.). For example, application sections on household composition, assets, or employment may require cultural sensitivity to be properly translated. Thus, the chatbot may fail to adequately capture the meaning and context of the questionnaire, thereby minimising the chances of the applicant's eligibility. In combination with the disproportionately high level of errors and hallucinations, asylum seekers from the Global South may receive an inferior level of assistance than their English-speaking counterparts. While the developer tried to mediate the impact of the language bias problem by using translation benchmarks to select the models, the systematic problem itself of course persists. This is especially concerning as the administrative context requires precision, with errors determining the outcome of applications and consequently affecting the economic situation of users. If left unaddressed, these errors may, in the worst case, increase the number of erroneous submissions, lengthen the application process, and potentially worsen the economic situation of some non-English-speaking users. However, due to the overall presence and necessity of machine translations in the field the use of LLM Chatbots likely merely reproduces rather than augments this risk, granting the caseworkers' awareness of automation bias.

6. Future Outlook & Recommendations

6.1 Future outlook

The project, having undergone a successful small-scale pilot phase, could be reinstated either within the same organisational setting or in another local chapter. Alternatively, public administrations themselves could experiment with similar tools to support beneficiaries more

broadly. Such systems could, for instance, be offered as an optional aid while still requiring beneficiaries to work with an assigned caseworker, thereby ensuring human oversight and accountability. Beyond use by non-German speakers, this type of tool could also benefit German-speaking applicants, as administrative language in official forms is often complex. LLMs are particularly well-suited to simplifying bureaucratic language and guiding users through the required questions in a clearer and more accessible manner.

More broadly, the possibility of reinstating or scaling the system must also be understood within the organisational culture of public and quasi-public institutions. Innovation in these environments tends to occur more easily at a small scale, often driven by individual initiative, whereas large-scale institutionalisation requires navigating complex governance structures, risk-averse cultures, and strict compliance requirements. These structural and cultural factors make it significantly more challenging to expand experimental tools like this chatbot beyond a pilot phase. Thus, any future deployment would not only depend on resolving legal uncertainties and securing resources, but also on organisational willingness to adopt and sustain innovative practices within the wider framework of public service provision.

6.2 Recommendations

1. Switch to Mistral API providers if feasible, due to cost considerations, data sovereignty and reduced privacy risks compared to US-based proprietary LLMs.
2. Develop comprehensive documentation, including explanations of the system's design choices, data handling procedures, and structured impact assessments to improve justifiability.
3. For the output delivered to the caseworker, display the entire conversation in both the applicant's language and German to enable transparent verification, facilitate contestability, and support discussions with clients when clarification is required.
4. Implement regular assessments of linguistic performance and bias, which include precision testing and monitoring potential systemic errors that may affect specific linguistic groups.

7. AI Disclaimer and Supporting Sources

7.1 AI-Disclaimer

AI was used to produce the flow chart with <https://codetoflow.com/>

No GenAI was used to produce any text; however, AI spell checkers, such as Grammarly, the Google Spell Checker, and Chat-GPT, were used for grammar and spelling corrections, as well as DeepL for translations.

7.2. General sources of law

General sources of law are cited in their latest form and can be accessed via the provided links. Specific articles are referenced using art. or §.

- Grundgesetz <https://www.gesetze-im-internet.de/gg/BJNR000010949.html>
- SGB <https://www.sozialgesetzbuch-sgb.de>
- GDPR <https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>
- VwVfG <https://www.gesetze-im-internet.de/vwvfg/>
- BVFG <https://www.gesetze-im-internet.de/bvfg/BJNR002010953.html>
- Asylblg <https://www.gesetze-im-internet.de/asylblg/>

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8. Methodological Annexe

8.1 Methodological considerations

The interview was conducted by three members of the group, one of whom served as the moderator, while the other two took notes, as the interview was not recorded due to anonymisation requirements. The developer agreed to participate in the study with the understanding that their name and the organisation for which they developed the chatbot would not be disclosed. The interview lasted over an hour and was conducted via the Zoom platform. For accountability purposes, this report will be sent to the developer. The interviewee was also provided with clear guarantees of confidentiality, including the secure handling of all notes and the assurance that no identifying information would be shared.

The interview was conducted using a semi-structured format, employing a flexible guideline that permitted adjustments as the interaction with the interviewee progressed. Following the interview, the notes collected by the observers were consolidated and collectively synthesised by the research team to ensure accuracy and a shared interpretation of the data.

A limitation of the process is that, due to anonymisation requirements, the interview could not be recorded, which may restrict the precision of direct quotations. Additionally, as the study relied on a single interview, the findings may not allow for methodological triangulation.

8.2 Interview Guidelines

Context & Development

- Can you describe the context in which the prototype was developed and explain why it was necessary? What problems were citizens or caseworkers facing with traditional *Bürgergeld* applications? Who were the key people or organisations guiding the vision?
- Walk me through the development process. Were there pilot phases or trials? What was the timeline from concept to deployment?

- Which elements were most challenging during development – for example, handling multiple languages, making the interface user-friendly, ensuring legal accuracy, or integrating with existing systems?
- How were different stakeholders involved in the design process – case workers, applicants, policy experts, or advocacy groups? Did their input significantly shape the system?

Technical Architecture & Data

- Which Gen-AI model are you using, and what influenced this choice? (Cost, data sovereignty, performance?) Is it a purely generative system, rule-based, or hybrid?
- What happens to users' personal data and conversation logs? Where is data stored and processed? Do you use conversations to retrain the model?
- What kind of human oversight is built into the process? At what points do humans intervene, review, or validate the chatbot's work?
- How does the system handle uncertainty, incorrect information, or edge cases? Are there confidence thresholds, human-in-the-loop mechanisms, or fallback procedures when the AI is unsure?
- Can you explain how the system translates legal/administrative language into plain language across different languages? Were certain languages or groups particularly challenging to support?
- Were you surprised by the user demographics – age range, digital literacy levels, or device usage patterns? What devices can people use to access the system?

Organisational Impact & Human Oversight

- How has this system changed the work of *Bürgergeld* caseworkers? What was their initial reaction – were there concerns about job displacement or deskilling? How do you handle disagreements between the chatbot's assessment and the caseworker's judgment?
- Has the system changed the volume, quality, or completeness of applications received? Any unintended consequences – positive or negative?

Policy Impact & Access

- Has the chatbot changed who applies for *Bürgergeld* or revealed anything about the policy's complexity? Have you seen demographic shifts or reduced barriers for specific communities (language groups, the elderly, etc.)?

- How does the system handle policy ambiguity or discretionary elements of *Bürgergeld* eligibility? Does it standardise decisions in ways that might affect outcomes compared to traditional caseworker discretion?

Governance, Transparency & Accountability

- How aware is the public of this tool? Was there any promotion, external guidance from organisations, or public backlash/controversies?
- What governance structure exists for oversight? How can users contest decisions influenced by the chatbot? What documentation or transparency mechanisms are in place (impact assessments, audit reports)?
- If the chatbot provides incorrect information that harms an applicant, who is responsible? How is accountability allocated between the AI system, developers, and the administration? What is the justification for outputs? If someone contests an outcome, who
- Is the system still in use? If not, why was it discontinued? If yes, what metrics indicate success or failure?

Conclusion

- If you could start over, what would you do differently? What recommendations would you give to other public administrations considering similar AI systems?
- What recommendations would you give to other public administrations considering similar AI systems?